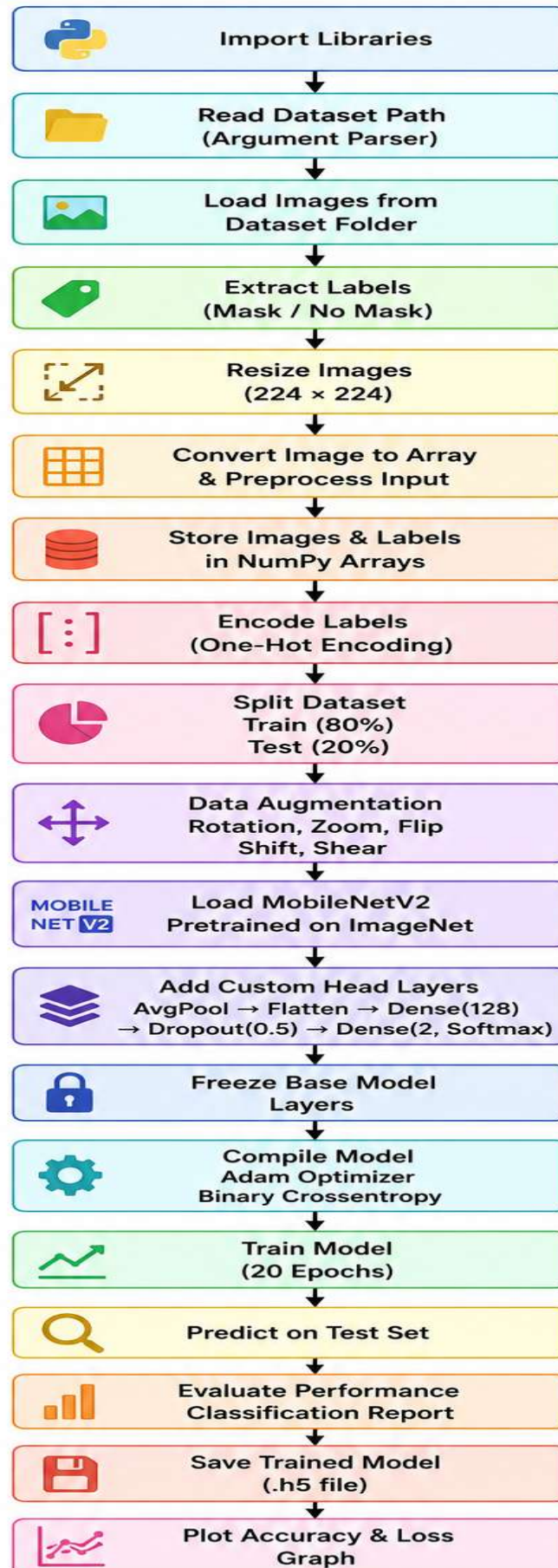
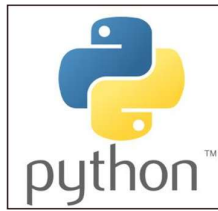


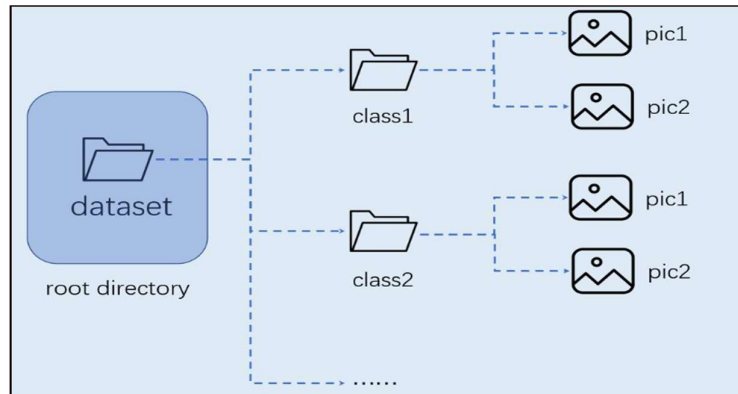
Detailed technical architecture



Import Libraries



Read Dataset Path



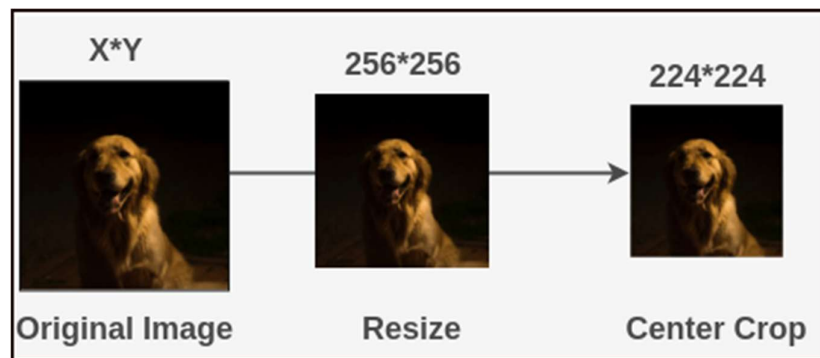
Load Images from Dataset Folder



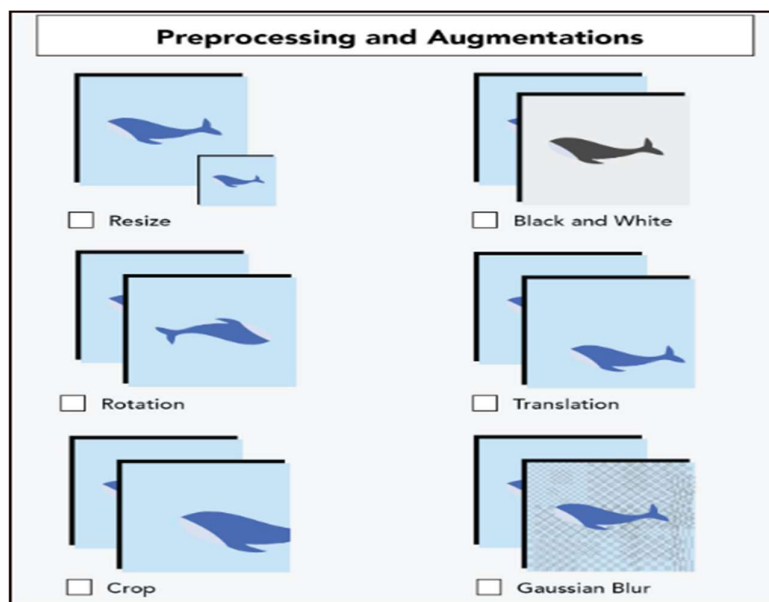
Extract Labels (Mask / No Mask)



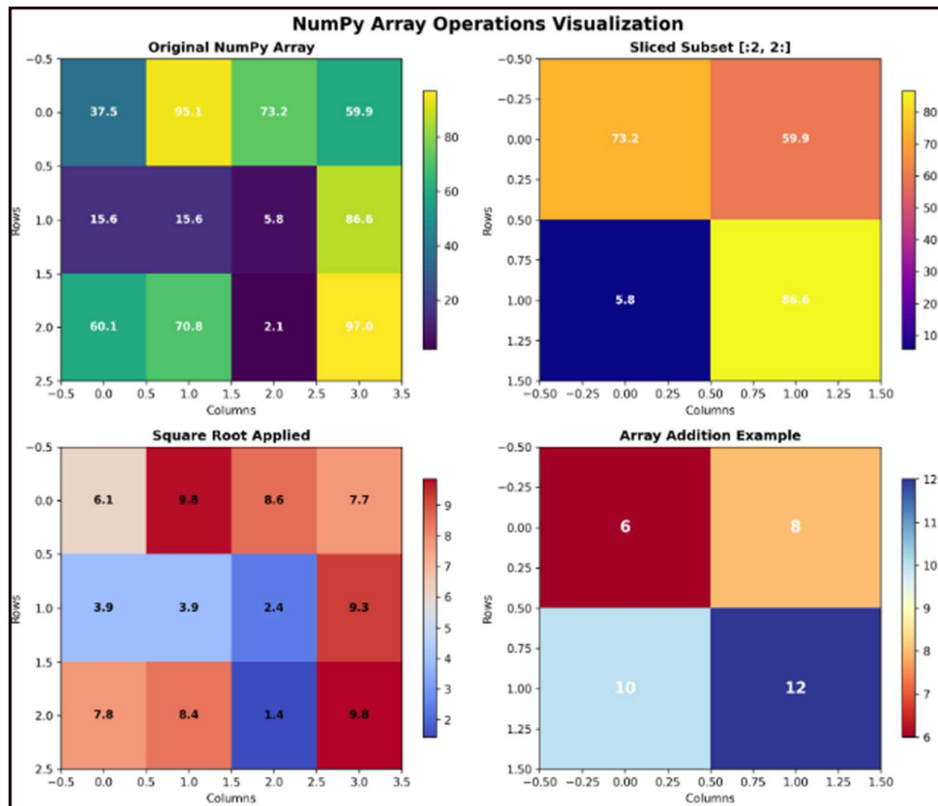
Resize Images (224×224)



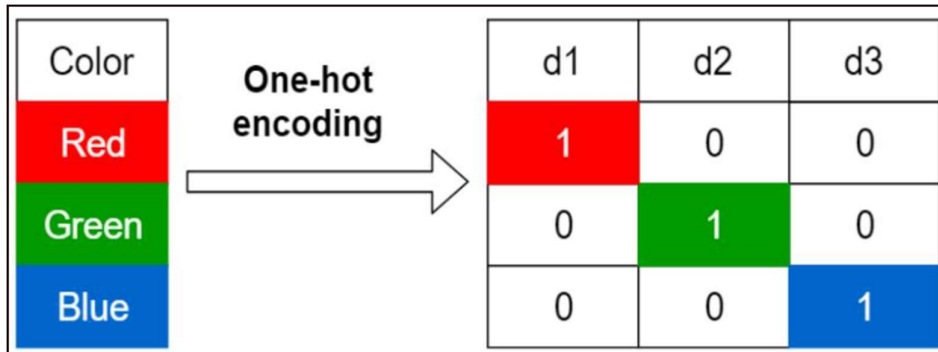
Convert Image to Array & Preprocess



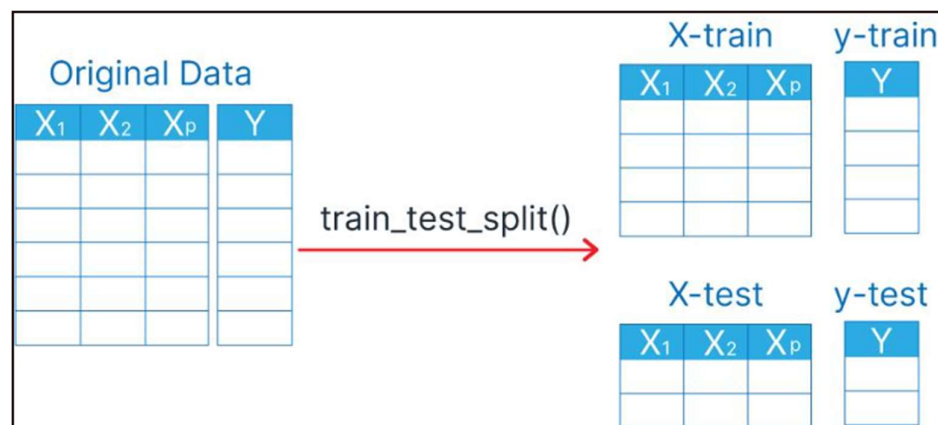
Store Images & Labels in NumPy Arrays



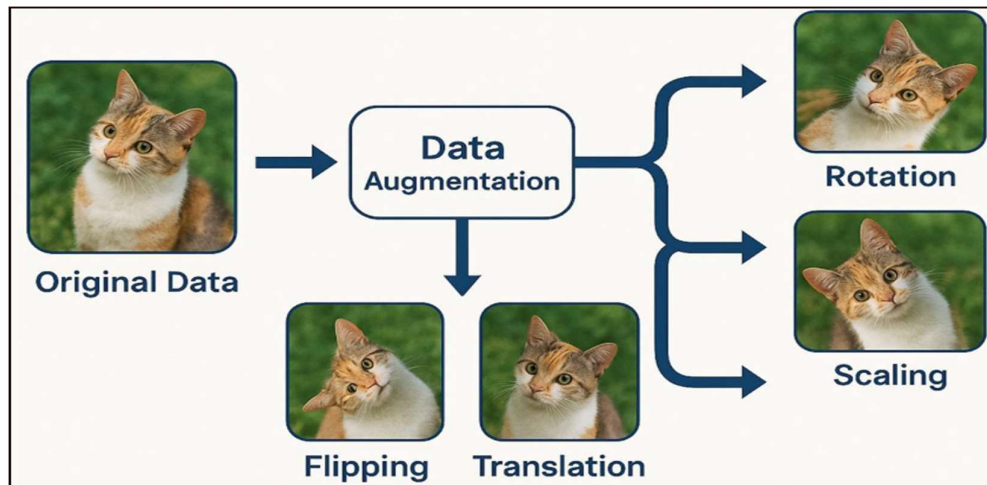
Encode Labels (One-Hot Encoding)



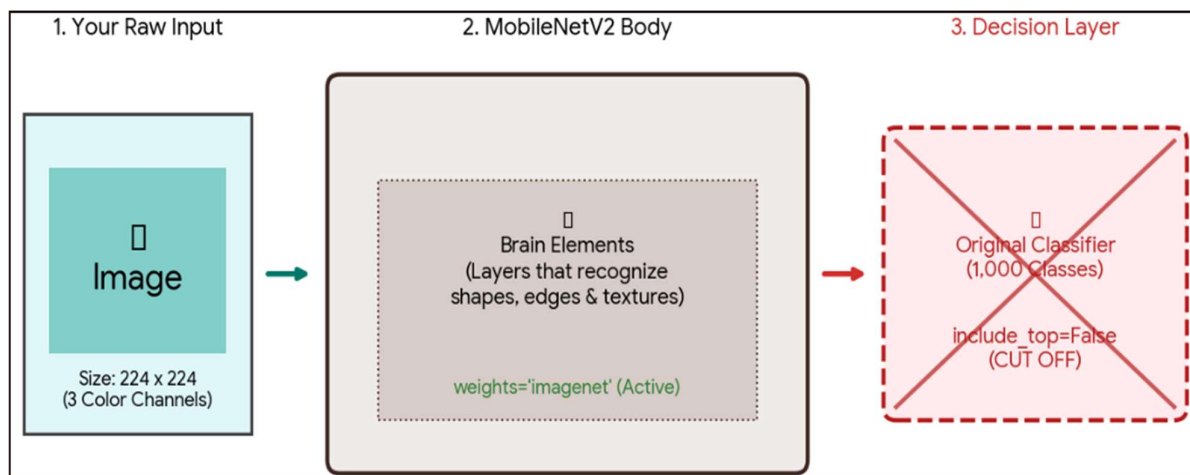
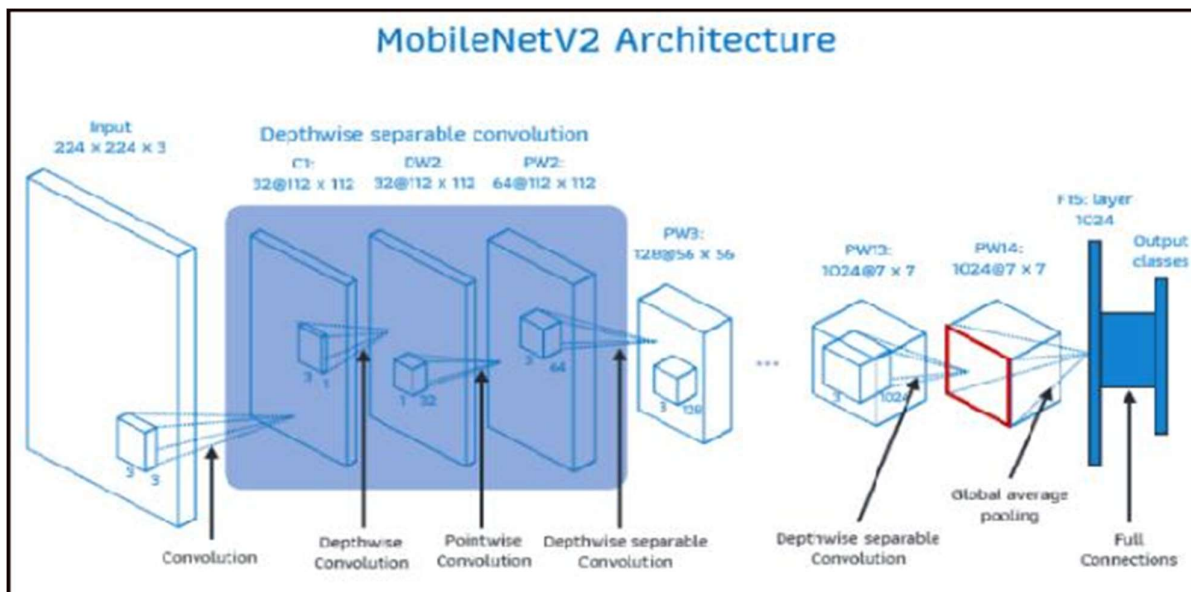
Split Dataset



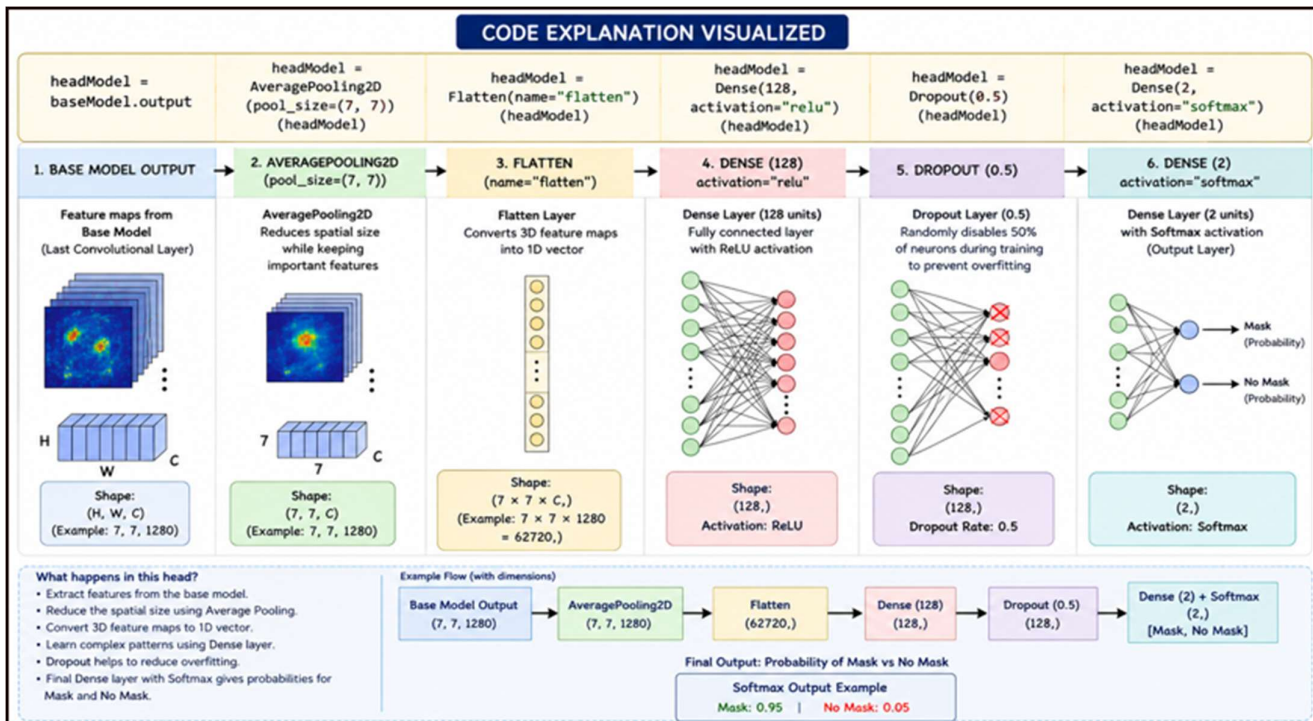
Data Augmentation



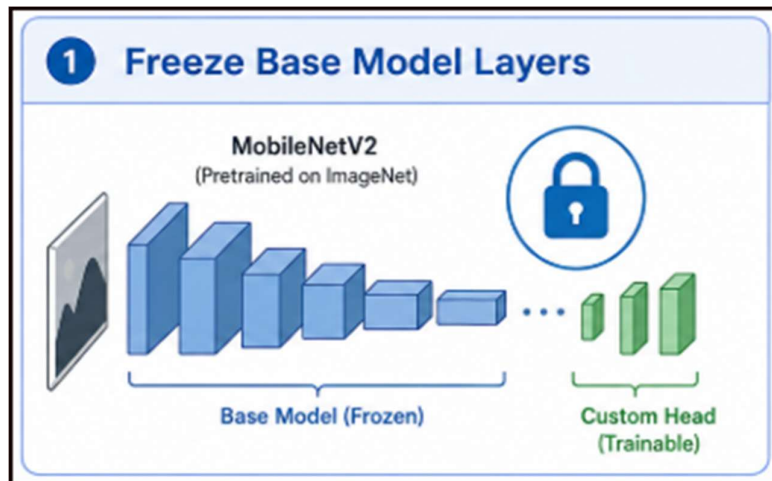
Load MobileNetV2(Transfer Learning)



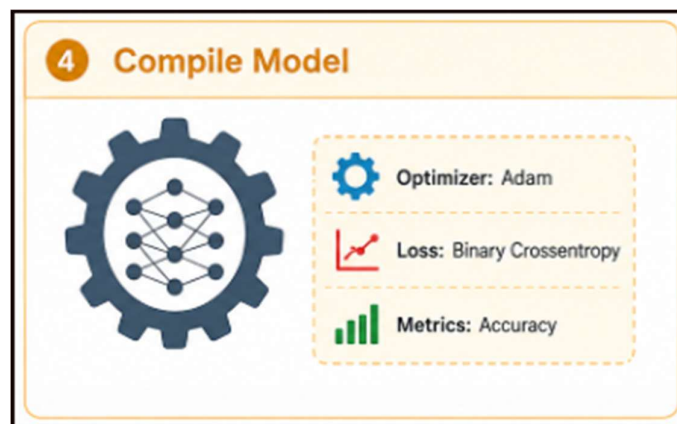
Add Custom Head Layers



Freeze Base Model Layers



Compile Model

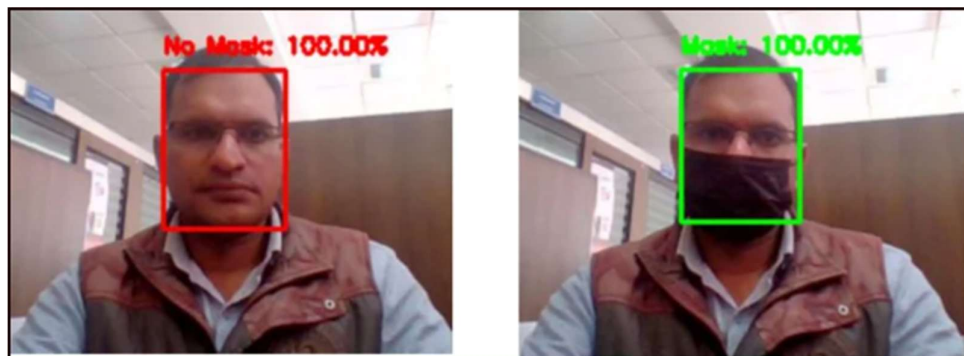


Train Model (20 Epochs)

Explanation of : `H = model.fit(aug.flow(trainX, trainY, batch_size=BS), steps_per_epoch=len(trainX) // BS, validation_data=(testX, testY), validation_steps=len(testX) // BS, epochs=EPOCHS)`

Part of Code	Explanation	Visual Representation
1 <code>aug.flow(trainX, trainY, batch_size=BS)</code>	- Creates a data generator using 'aug' (data augmentation). - Takes training images (trainX) and labels (trainY) and generates batches of size 'BS' for training.	Training Images (trainX) Data Augmentation (aug) - Rotation - Shift - Zoom - Flip ... Batches of Size BS
2 <code>steps_per_epoch=len(trainX) // BS</code>	- Number of batches to draw from the generator in one epoch. <code>len(trainX)</code> = total number of training samples. '/' gives integer (floor) division. - Ensures each epoch sees all training data once.	Total Training Samples <code>len(trainX) = N</code> Batch 1 (BS samples) → Batch 2 (BS samples) → Batch 3 (BS samples) → ... → Batch N//BS (BS samples) Total Batches per Epoch = $N // BS$
3 <code>validation_data=(testX, testY)</code>	- Provides validation (test) data to evaluate the model after each epoch. <code>testX</code> = validation images <code>testY</code> = validation labels	Validation Data testX (Validation Images) + testY (Validation Labels) 0 1 0 1 ... (Corresponding Labels)
4 <code>validation_steps=len(testX) // BS</code>	- Number of batches from the validation data used to evaluate the model. <code>len(testX)</code> = total number of validation samples. '/' gives integer (floor) division.	Total Validation Samples <code>len(testX) = M</code> Batch 1 (BS samples) → Batch 2 (BS samples) → Batch 3 (BS samples) → ... → Batch M//BS (BS samples) Total Validation Batches = $M // BS$
5 <code>epochs=EPOCHS</code>	Total number of times the model will see the entire training data (one complete pass through the dataset).	Training for EPOCHS epochs Epoch 1 (Trains on all batches (1 to N//BS)) → Epoch 2 (Trains on all batches (1 to N//BS)) → Epoch 3 (Trains on all batches (1 to N//BS)) → ... → Epoch EPOCHS (Trains on all batches (1 to N//BS))
In Simple Words	This line trains the model using augmented training data in batches, validates it on the test data after each epoch, and repeats the whole process for a total of EPOCHS epochs.	

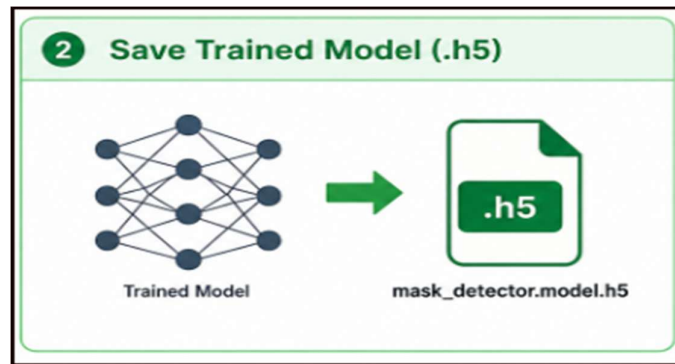
Predict on Test Set



Evaluate Performance

	precision	recall	f1-score	support
0	0.77	0.86	0.81	37584
1	0.84	0.75	0.79	37577
accuracy			0.80	75161
macro avg	0.81	0.80	0.80	75161
weighted avg	0.81	0.80	0.80	75161

Save Trained Model (.h5)



Plot Accuracy & Loss Graph:

